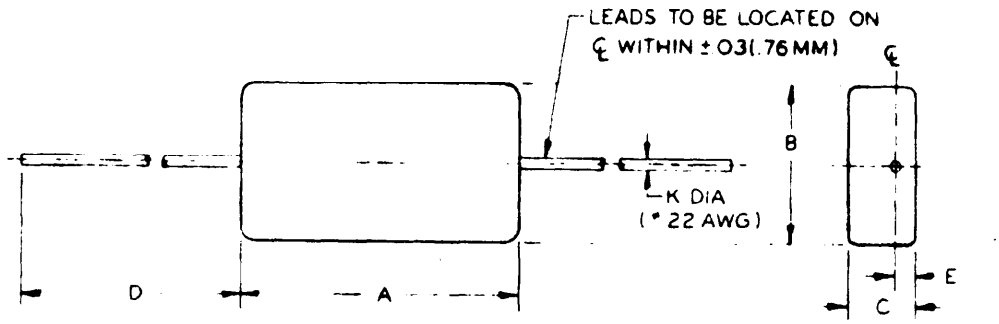
MIL-C-5/1F
12 AUGUST 1966
SUPERSEDING
MIL-C-5/1E
28 September 1961

MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, MICA DIELECTRIC,

STYLE CM15

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-5.



DIMENSIONS						
CIM	INCHES			MILLIMETERS		
	MIN	NOM	MAX	MIN	NOM	MAX
A	.48	.52	.55	12.19	13.21	13.97
B	.27	.30	.31	6.86	7.62	7.87
C	.14	.19	.22	3.56	4.83	5.59
D	1.12	---	---	28.45	---	---
E	.06	---	---	1.52	---	---
K	.023	.025	.027	.58	.64	.69

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Approximate weight .05 ounce.
4. Unless otherwise specified, tolerance is .xx ±.02.

FSC 5910

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REQUIREMENTS:

Dimensions: See figure.

Capacitance value: See table I.

Capacitance tolerance: See table I.

Symbol	Capacitance tolerance
D	±.5 pf
F	±1 percent
G	±2 percent
J	±5 percent

DC rated voltage: See table I.

Operating temperature range: See table I.

Symbol	Temperature range
N	-55°C to +85°C
O	-55°C to +125°C
P	-55°C to +150°C

Vibration grade: 3 (10 to 2, 000 cps).

Characteristic: See table I.

Symbol	Temperature coefficient P/M/°C	Capacitance drift
C	-200 to +200	±(0.5 percent + 0.1 pf)
E	-20 to +100	±(0.1 percent + 0.1 pf)
F	0 to +70	±(0.05 percent + 0.1 pf)

Dielectric withstanding voltage:
Method 301 of MIL-STD-202:
200 percent of rated dc voltage for not less than 1 second, nor more than 5 seconds.

Barometric pressure:
Method 105 of MIL-STD-202: Condition D (100, 000 ft)
Test potential - 150 percent of rated dc voltage for not less than 1 second nor more than 5 seconds.

Insulation resistance:
Method 302 of MIL-STD-202, condition A.
At 25°C - 100, 000 megohms minimum.
At 85°C - 15, 000 megohms minimum.
At 125°C - 10, 000 megohms minimum.
At 150°C - 5, 000 megohms minimum.

Dissipation factor, maximum: See MIL-C-5.

Capacitance:
Method 305 of MIL-STD-202.

Vibration:
Method 204 of MIL-STD-202, condition D.

Temperature cycling:
Method 102 of MIL-STD-202, condition D.

Immersion:
Method 104 of MIL-STD-202, condition B.
Dielectric withstanding voltage - Meet initial requirement.
Insulation resistance - At least 30 percent of initial requirement.
Dissipation factor - Not greater than 150 percent of initial requirement.
Capacitance change - Shall not exceed ±2 percent of nominal value or 1 pf, whichever is greater.

Shock (medium impact):
Method 205 of MIL-STD-202, condition C (50G).

Solderability:
Method 208 of MIL-STD-202.

Terminal strength:
Pull test - Capacitor held by one terminal and 5 pound load gradually applied to other terminal and maintained for at least 5 seconds.
Twist test - 5 alternate 360° rotations.

Moisture resistance:
Method 106 of MIL-STD-202.
Dielectric withstanding voltage - Initial requirement.
Insulation resistance - At least 30 percent of initial requirement.
Dissipation factor - Not greater than 150 percent of initial requirement.
Capacitance change - Shall not exceed ±2 percent of nominal value or 1 pf, whichever is greater.

Temperature coefficient and capacitance drift:
Within the limits specified for the applicable characteristic.

Low temperature storage and life:
48 hours at $-55\ ^{+0}_{-2}$ °C, immediately followed by qualification or performance test, as applicable.
Qualification test - 2, 000 hours at applicable high test temperature with 150 percent of rated dc working voltage applied.
Dielectric withstanding voltage - Initial requirement.
Insulation resistance - Initial requirement.
Dissipation factor - Not greater than 150 percent of initial requirement.
Capacitance change - Shall not exceed ±2 percent of nominal value or 1 pf, whichever is greater.
Performance check - Conditions and requirements in accordance with qualification test except test duration shall be 250 hours.

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TABLE I. Style CM15.

Type Designation 1/	Capacitance (pf)	DC working voltage (volts)	Operating temperature range	Characteristics available for capacitance tolerances			
				± .5 pf (D)	± 1% (F)	± 2% (G)	± 5% (J)
CM15CD010D-3	1	500	N, O, P	C			
CM15CD020D-3	2	500	N, O, P	C			
CM15CD030D-3	3	500	N, O, P	C			
CM15CD050D-3	5	500	N, O, P	C			
CM15CD100J-3	10	500	N, O, P				C
CM15CD110J-3	11	500	N, O, P				C
CM15CD120J-3	12	500	N, O, P				C
CM15CD130J-3	13	500	N, O, P				C
CM15CD150J-3	15	500	N, O, P				C
CM15CD160J-3	16	500	N, O, P				C
CM15CD180J-3	18	500	N, O, P				C
CM15CD200J-3	20	500	N, O, P				C
CM15CD220J-3	22	500	N, O, P				C
CM15CD240J-3	24	500	N, O, P				C
CM15-D270--3	27	500	N, O, P			CE	CE
CM15-D300--3	30	500	N, O, P			CE	CE
CM15-D330--3	33	500	N, O, P			CE	CE
CM15-D360--3	36	500	N, O, P			CE	CE
CM15-D390--3	39	500	N, O, P			CE	CE
CM15-D430--3	43	500	N, O, P			CE	CE
CM15-D470--3	47	500	N, O, P			CE	CE
CM15-D510--3	51	500	N, O, P		CEF	CEF	CEF
CM15-D560--3	56	500	N, O, P		CEF	CEF	CEF
CM15-D620--3	62	500	N, O, P		CEF	CEF	CEF
CM15-D680--3	68	500	N, O, P		CEF	CEF	CEF
CM15-D750--3	75	500	N, O, P		CEF	CEF	CEF
CM15-D820--3	82	500	N, O, P		CEF	CEF	CEF
CM15-D910--3	91	500	N, O, P		CEF	CEF	CEF
CM15-D101--3	100	500	N, O, P		CEF	CEF	CEF
CM15-D111--3	110	500	N, O, P		CEF	CEF	CEF
CM15-D121--3	120	500	N, O, P		CEF	CEF	CEF
CM15-D131--3	130	500	N, O, P		CEF	CEF	CEF
CM15-D151--3	150	500	N, O, P		CEF	CEF	CEF
CM15-D161--3	160	500	N, O, P		CEF	CEF	CEF
CM15-D181--3	180	500	N, O, P		CEF	CEF	CEF
CM15-D201--3	200	500	N, O, P		CEF	CEF	CEF
CM15-D221--3	220	500	N, O, P		CEF	CEF	CEF
CM15-D241--3	240	500	N, O, P		CEF	CEF	CEF
CM15-D271--3	270	500	N, O, P		CEF	CEF	CEF
CM15-D301--3	300	500	N, O, P		CEF	CEF	CEF
CM15-D331--3	330	500	N, O, P		CEF	CEF	CEF
CM15-D361--3	360	500	N, O, P		CEF	CEF	CEF
CM15-D391--3	390	500	N, O, P		CEF	CEF	CEF
CM15-D431--3	430	500	N, O, P		CEF	CEF	CEF
CM15-D471--3	470	500	N, O, P		CEF	CEF	CEF
CM15-D511--3	510	500	N, O, P		CEF	CEF	CEF

1/ Complete type designation where applicable, will include additional symbols to indicate characteristic, capacitance tolerance, and temperature range.

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Continuation test - Units that have completed the 250 hour performance shall be continued for an additional 1750 hours and meet the requirements for the qualification test.

Qualification inspection: Qualification inspection (Group A) is required.

Supersession data: Capacitors as described herein with capacitance values ranging from 51 through 510 pf, may be considered as suitable replacements for comparable items formerly covered by MIL-C-5/3C. Recommended replacements are as follows:

MIL-C-5/3C items	Recommended military replacement items
CM25E510GN3 thru CM25E511GN3	CM15ED510GN3 thru CM15ED511GN3
CM25E510GO3 thru CM25E511GO3	CM15ED510GO3 thru CM15ED511GO3
CM25E510GP3 thru CM25E511GP3	CM15ED510GP3 thru CM15ED511GP3
CM25E510JN3 thru CM25E511JN3	CM15ED510JN3 thru CM15ED511JN3
CM25E510JO3 thru CM25E511JO3	CM15ED510JO3 thru CM15ED511JO3
CM25E510JP3 thru CM25E511JP3	CM15ED510JP3 thru CM15ED511JP3
CM25F910GN3 thru CM25F511GN3	CM15FD910GN3 thru CM15FD511GN3
CM25F910GO3 thru CM25F511GO3	CM15FD910GO3 thru CM15FD511GO3
CM25F910GP3 thru CM25F511GP3	CM15FD910GP3 thru CM15FD511GP3
CM25F910JN3 thru CM25F511JN3	CM15FD910JN3 thru CM15FD511JN3
CM25F910JO3 thru CM25F511JO3	CM15FD910JO3 thru CM15FD511JO3
CM25F910JP3 thru CM25F511JP3	CM15FD910JP3 thru CM15FD511JP3

Custodians:
Army - EL
Navy - SH
Air Force - 11

Review activities:
Army - EL
Navy - SH
Air Force - 11, 17, 85

User activities:
Army -
Navy -
Air Force - 14, 19

Preparing activity:
Army - EL

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5. PROBLEM AREAS

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c. Reason/Rationale for Recommendation:

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